A Condition-Indexed Operating Envelope and Representation-Repair Study for Automatic Modulation Classification under Structured Interference

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***Abstract*—Uniform model rankings hide a central design question in automatic modulation classification (AMC): under structured interference, the ranking of a compact spectral representation and a much larger I/Q-domain classifier can itself depend on physical conditions. We study this rank exchange in source-disjoint 3GPP TR 38.901 TDL-profile simulations, comparing VIMD-Net, a 39500-parameter spectral network, with strong I/Q-domain comparators across the complete SNR–SIR plane. The sealed confirmatory family resolves a positive whole-configuration contrast of 4.57 percentage points ($[3.65, 5.49]$ percentage points) over its shared spectral backbone, while the two component-level interventions are bounded rather than individually resolved. Marginally over the hard-interference split the compact model trails both I/Q comparators (macro-F1 0.4406 versus 0.4599 and 0.4872), but the ranking reverses in the severe-interference region: at SIR $= -15$ dB the compact model gains 9.74 percentage points over MCLDNN and 10.46 percentage points over IQFormer-inspired. A two-variable overlap–SIR envelope, fitted on in-distribution and hard-interference cells only, predicts the held-regime gain against IQFormer-inspired with RMSE 6.23 percentage points versus 8.37 percentage points for a fit-domain constant ($R^2_{\text{skill}} = 0.45$). A failed clean-retention gate is localized by frozen probes and narrowed by a received-I/Q sidecar that raises clean macro-F1 by 11.77 percentage points; these Tier-2 results are exploratory and outside the sealed family. The paper reports a simulation-only operating envelope and a partial representation-repair path, not uniform superiority.**

***Index Terms*—Automatic modulation classification, structured interference, interference overlap, vehicular Doppler, operating envelope, condition-dependent model comparison, representation repair, compact neural networks.**

Evidence status. The original sealed family resolves one positive whole-configuration contrast (A5 vs A0), while two component contrasts and the clean-retention gate do not pass. Severe-interference, envelope, and Tier-2 repair results are exploratory or post-hoc; all conclusions are restricted to source-disjoint simulation.

I. Introduction

Automatic modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform's modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. A high-mobility terminal facing structured interference must decide on a limited compute budget whether to run a compact spectral front end or a high-capacity I/Q path; the same model can trail the high-capacity I/Q classifier in weak or narrowband interference yet lead it in a severe corner where the interference overlaps the target's own time–frequency support. Averaging these regimes produces a ranking but erases the design rule.

This paper asks a narrower question: *can the sign and size of the compact-model advantage be indexed by physical condition variables, and can the boundary revealed by that index be repaired?* We answer using a frozen source-disjoint simulation campaign, artifact-only post-hoc analysis, and prospective Tier-2 experiments. The base model, VIMD-Net, maps a complex STFT to target-dominant, jammer-dominant, and unexplained-or-ambiguous routes; its supervision is physics-informed in the sense that a simulation-component teacher, available only during training, is derived from independently tracked target, jammer, and residual components. We do not claim that a specific routing or teacher component is individually confirmed (Section V-A); the contribution is the boundary and its repair, not the architecture itself.

The contributions are:

- a condition-indexed rank exchange, exposed over the complete structured-interference SNR–SIR plane, between a compact spectral model and stronger I/Q-domain classifiers, instead of a single marginal ranking;
- a low-dimensional target-support-overlap–SIR empirical envelope whose cross-regime gain-transport skill is evaluated on regime cells excluded from fitting, against an explicit fit-domain constant baseline and with a bounded, non-causal interpretation;
- a separation of the original sealed whole-configuration contrast from post-hoc severe-interference evidence, using paired statistics, multiplicity control, and full-region visualization so a favorable corner is not read as uniform superiority; and
- a prospective diagnosis-and-repair chain that turns a failed clean-retention gate into a representation-information finding: a received-I/Q sidecar substantially narrows the clean failure while a spectral capacity ladder shows that width alone does not remove the boundary.

The claim is deliberately not that VIMD-Net is uniformly best. It is that the compact and high-capacity representations exchange rank in a reproducible, condition-dependent way, that physical covariates can describe this boundary, and that, for its principal clean boundary, the evidence is more consistent with a repairable representation-information gap than with insufficient clean exposure alone.

II. Related Work and Claim Boundary

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated reparameterized causal convolutions, cross-scale kernel selection with IQ-imbalance and carrier-offset compensation, and channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust, Cross-Domain, and Interference-Aware AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16], while overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation populate the 2024–2026 frontier [17]–[19]. Interference-tolerant recognition [16] and tensor-based malicious-interference recognition for MIMO [20] address structured interference directly. Our setting differs: we neither reconstruct sources nor seek universal interference robustness; we study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

C. Distinction: Condition-Indexed Ranking and Repair

Knowledge-guided and multimodal preprints continue these directions [21]–[24]. Our contribution is not another average-dominance claim: we model the condition dependence of the ranking itself, fit a low-dimensional boundary to that rank exchange, expose the negative regions, and prospectively intervene on the revealed boundary—a positioning claim, not an exhaustive systematic review.

III. System Model and VIMD Configuration

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa\mathcal{H}_u(u)[n] + w[n])\,, \tag{1}$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [25] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10\log_{10}\big(E[|\mathcal{H}_s(s)|^2]/E[|\kappa\mathcal{H}_u(u)|^2]\big)$ on the simulator's tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [26], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. The held-speed split probes deeper intra-window non-stationarity: $T_c \approx 0.423/f_D$ falls to about 0.31 ms at 250 km/h, below the 1024 μs decision window.

Training pairs two independently impaired observations of the same immutable source; inference uses one view, so pairing is a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, swapping two admissible emitters can preserve the mixture while changing the requested label.

B. Three-Route Spectral Allocation

Let $Z = \mathrm{STFT}(x) \in \mathbb{C}^{F\times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned}[M_s, M_j, M_o] &= \mathrm{softmax}([A_s, A_j, A_o]),\\ M_s + M_j + M_o &= 1.\end{aligned} \tag{2}$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ. The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{3}$$

$$Z_j = (M_j + (1-\lambda)M_o) \odot Z. \tag{4}$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range $[0.05, 0.35]$. It is added only to the modulation route; therefore the retained route weights remain nonnegative but sum to $1+\rho$ rather than one, and the allocation is no longer an exact partition of the mixture power, i.e. $Z_m + Z_j \neq Z$ in general. This is the ratio-masking construction familiar from supervised source separation [27]–[30]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^\star = [q_s - q_j]_+, \qquad M_j^\star = [q_j - q_s]_+, \tag{5}$$

$$M_o^\star = q_u + 2\min(q_s, q_j). \tag{6}$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

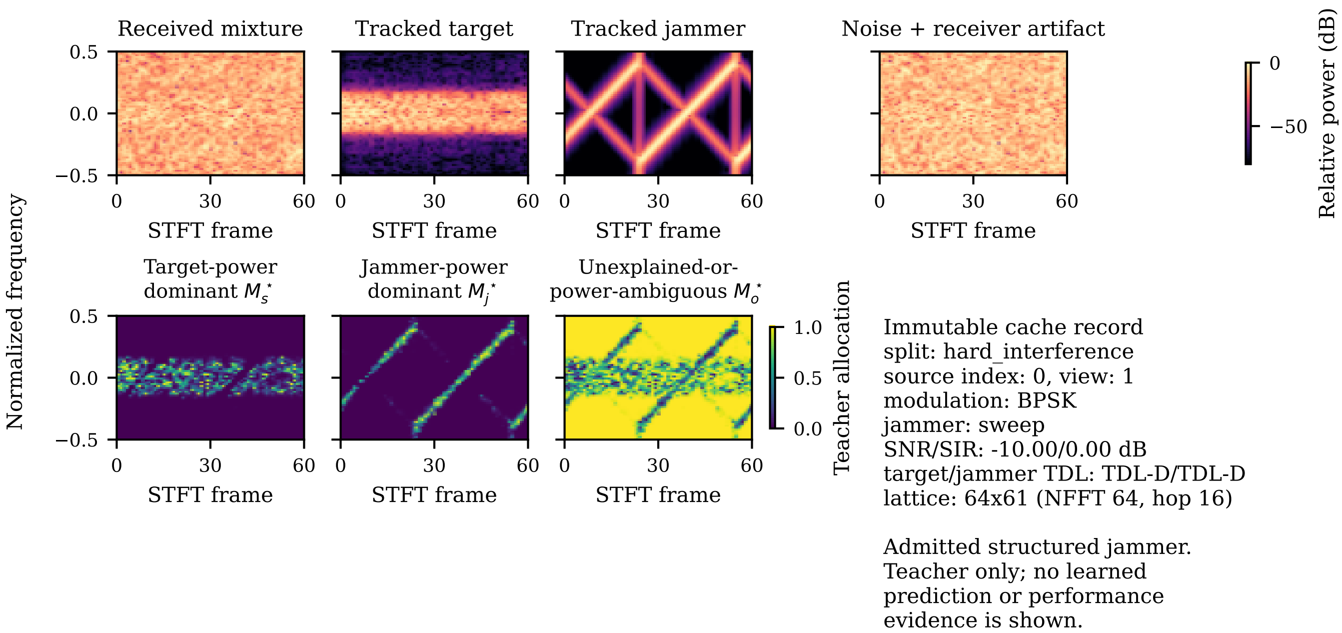

Fig. 1. A deterministic cache record and the fixed simulation-component teacher of Eqs. (5)–(6). Top row: received mixture, tracked target, tracked jammer, and noise-plus-artifact spectrograms; bottom row: the target-dominant, jammer-dominant, and uncertainty route masks. The teacher is a training-only construction from simulator-tracked components; it is unavailable at inference and is not performance evidence.

D. Training Objective

The total loss is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\mathrm{mod}} + \alpha\mathcal{L}_{\mathrm{teacher}} + \beta\mathcal{L}_{\mathrm{jammer}} + \gamma\mathcal{L}_{\mathrm{link}} \\ + \eta\mathcal{L}_{\mathrm{contrast}} + \theta\mathcal{L}_{\mathrm{orth}},\end{aligned} \tag{7}$$

where $\mathcal{L}_{\mathrm{mod}}$ is the modulation classification loss, $\mathcal{L}_{\mathrm{teacher}}$ is a Jensen–Shannon [31] alignment between student and teacher masks, $\mathcal{L}_{\mathrm{jammer}}$ and $\mathcal{L}_{\mathrm{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\mathrm{contrast}}$ is a supervised-contrastive loss [32] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\mathrm{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective, not a headline contribution.

E. Information Contract and Complexity

At inference every model receives only the mixture; simulator-tracked target, jammer, and SIR quantities are training-only or post-hoc audit variables. VIMD-Net uses 39500 parameters and 41.8 million reported MACs.

F. Implementation Details

Unless otherwise stated, all models share the same source-disjoint cache, data budget, optimization budget, stopping rule, and validation-only selection criterion; model-specific components follow each variant's predefined configuration.

Input and front end. One example is a complex baseband window of 1024 samples (1024 μs at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control. The spectral front end applies a two-sided STFT with $N_{\mathrm{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospective capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4}, weight decay 10^{-2}, batch size 16, mixed precision, gradient-norm clipping at 5.0, and a shared 30-epoch budget. Auxiliary losses follow a fixed staged schedule, and the learning rate decays on a cosine to a floor of 0.05 times its initial value once the staged schedule is complete. The reported checkpoint is chosen solely by validation modulation cross-entropy under a common eligibility window and 8-epoch early-stopping rule. Optimizer settings, schedule, and selection rule are identical for every model; no comparator received an architecture-specific sweep, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 bundle but not isolated.

Seeds. The sealed campaign uses 10 algorithm seeds shared across all models, with seed-aligned initialization; matched-subset displays use the first five. Exact schedules, seed identifiers, teacher construction, and cache provenance are recorded in the reproduction package. Fig. 1 is a method illustration, not performance evidence.

IV. Evidence Protocol

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans SIR $\in \{-15, -10, -5, 0\}$ dB, so SIR $= -15$ dB occurs only in the hard-interference split and SIR $= +5$ and $+10$ dB only in the in-distribution split. Each hard-interference SNR–SIR evaluation cell of Fig. 2 contains 131–181 source-disjoint test windows (mean 156; all ten classes, 5–26 per class), read from the frozen prediction cache. The envelope aggregates of Section V-C use a coarser partition of the same predictions (216–279 source-disjoint windows per held cell, mean 250). The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package but does not enter any sealed, envelope, or reported model-selection claim.

B. Comparator Fairness

Table I summarizes the comparators. All models are retrained under the same source-disjoint protocol; "IQFormer-inspired" and "MCLDNN" denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as an adaptation-oriented reference rather than a tuned upper bound; its 8.63 million parameters but only 155.3 million MACs reflect aggressive encoder downsampling and a large projection head, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator "IQFormer-insp."; "A5" and "VIMD-Net" denote the same sealed configuration, and "CSSL" abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [33] we retrain under our protocol. Because this campaign retrains CSSL under the common supervised budget without executing the released self-supervised pretraining stage, it is reported as a supervised adaptation reference rather than a reproduction of the full released recipe.

As a public-benchmark comparator-fidelity check, the MCLDNN and IQFormer implementations were additionally evaluated on the public RML2016.10a benchmark [1] under the protocol of the official benchmark result table of the IQFormer release [34]: the 11-class set, the full -20 to 18 dB SNR grid, per-stratum 600/200/200 splits seeded at 233, no augmentation or input-length change. Over three algorithm seeds, all-SNR test accuracy is 58.55% for MCLDNN (seed range 58.1–59.4%) and 65.23% for IQFormer-inspired (64.8–65.8%; seed ranges, not confidence intervals); the reference-release values, averaged over the same twenty SNR bins, are 62.05% and 64.19%. IQFormer-inspired therefore closely matches the reference-release all-SNR accuracy (+1.04 pp), whereas MCLDNN remains moderately lower (-3.50 pp) while staying in the same broad all-SNR range. Per-SNR deviations of up to about 8 pp remain, so E1 is range-level comparator-implementation fidelity, not pointwise reproduction nor evidence for the simulator-domain envelope.

TABLE I
Comparator and Complexity Summary (Hard-Interference Setting).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A5 differs from A0 in allocation, teacher, auxiliary objectives, residual bypass, and capacity (about 4.7× parameters), so A5–A0 is a whole-configuration effect.

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, overlap, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was preregistered separately and did not pass; clean behavior is therefore treated as a boundary to explain and repair, not as a preserved property of the sealed model.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, an exploratory calibration rather than a sealed significance threshold. Because seeds are

TABLE II
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	−0.76	1.08
Tri-route vs dual-route	0.44	−0.47	1.36

fixed blocks, the reported intervals are conditional on the realized seed set; a seed-randomization variant resampling the ten realized seeds with replacement keeps the A5–A0 contrast positive (95% percentile interval $[4.02, 5.19]$ percentage points) while the teacher-form and route-count contrasts span zero ($[-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points). In the severe corner, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples. The SIR $= -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2.

V. RESULTS

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table II reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points, about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their simultaneous upper bounds are 1.08 and 1.36 percentage points, respectively. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three contrasts, the family gate did not pass. Because the A5–A0 interval is part of the same simultaneous family-wise band, its positive whole-configuration contrast remains a valid family-wise-controlled result even though the prespecified joint family gate does not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 4); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design's power; we therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to any single component.

B. Rank Exchange across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models, but Fig. 2 shows why no single ranking is adequate: rank changes with SNR and SIR, with A5 reversing the marginal ranking under severe interference; the overlap analysis in Section V-C further shows that the required overlap decreases as interference becomes more severe.

At SIR $= -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ($[7.52, 12.14]$) and 10.46 percentage points over IQFormer-inspired ($[6.78, 14.29]$); these two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks reinforce the corner result without converting the post-hoc analysis into confirmation.

C. Empirical Overlap–SIR Operating Envelope

The rank exchange motivates an empirical envelope fit rather than another marginal average. For reference b, we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\,\mathrm{SIR}, \tag{8}$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*. We avoid the more common "spectral occupancy" because o is not an occupied-bin or occupied-bandwidth fraction. The covariate is defined from the simulator-tracked target and jammer components rather than from the mixture. For each window we compute the STFT power of the tracked target and of the tracked jammer separately, order the time–frequency cells by descending target power, and take the target's 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer's total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t)\in\mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \tag{9}$$

Thus $o = 0.8$ means that 80% of the jammer's energy lies within the target's dominant support, not that the jammer fills 80% of the plane. The cell-level covariate is the mean of o over the windows of that cell.

Equation (9) is evaluated on a fixed analysis lattice, a Hann-windowed 128-point STFT with hop 32, deliberately independent of the 64-point VIMD front-end lattice of Section III-F: the front-end lattice is what the model sees, whereas the analysis lattice is a measurement instrument for the physical audit. The overlap is computed once per window when the cache is built, before any model is trained; it is stored in the sealed cache, checksummed in the cache manifest, and read unchanged by every model, split, and comparison in this paper. It is never an input feature of VIMD-Net or of any comparator, and it is not tuned against any model's predictions.

The fit uses 32 aggregates, reconstructible from the protocol rather than asserted. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten classes. The `hard_interference` split contributes 4 SIR strata $(-15, -10, -5, 0\text{ dB}) \times 4$ overlap quartiles $= 16$ cells, and `id_test` contributes 16 further cells over its five SIR strata; its lowest-overlap quartile is dominated by jammer-free windows ($o = 0$, SIR $= 0$ dB cache sentinel) and clears the window floor only at 0 dB, which fixes the per-stratum counts. The primary held set is the 80 finite-SIR interfered

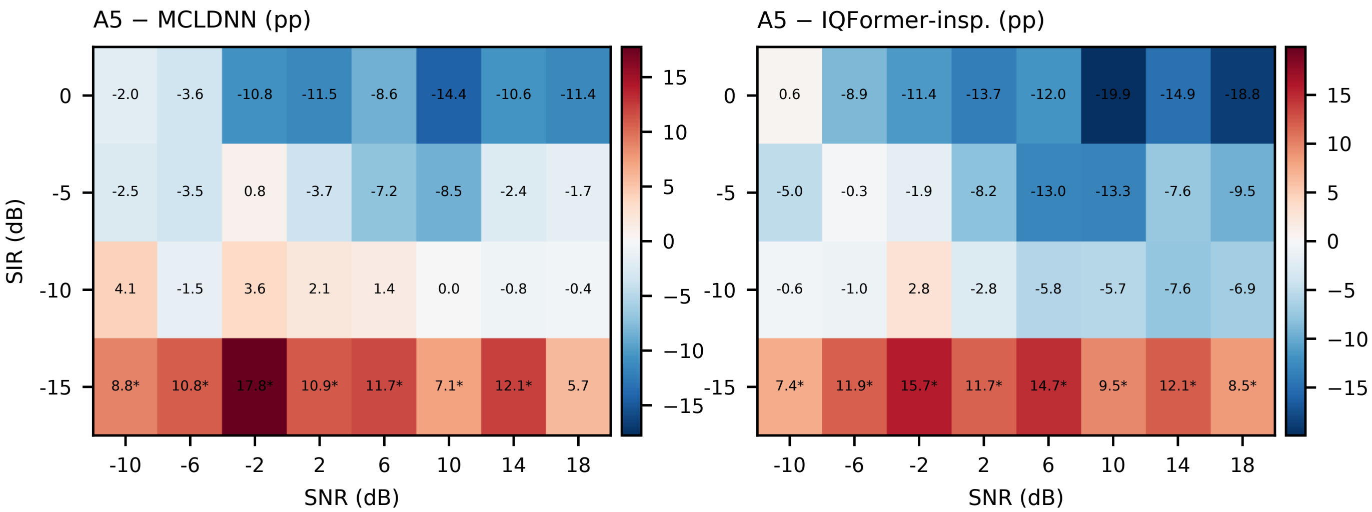

Fig. 2. Post-hoc operating map over all hard-interference SNR–SIR cells; each entry is A5 minus the named strong baseline in macro-F1 percentage points. A star marks a positive unadjusted paired lower interval; Holm- surviving cells are reported in the text. Displaying the full map prevents the severe corner from being read as a uniform ranking.

cells formed the same way from unseen jammer, unseen speed, held-out channel, and combined OOD (4×5 SIR strata $\times$ 4 quartiles $= 80$). Clean retention is deliberately outside this primary envelope because a jammer-free window has no physical finite SIR. Overlap and SIR are nearly orthogonal across the fitting cells (Pearson $r = -0.003$; -0.03 over all 113 cells), so strong linear collinearity between the two fitted covariates is not evident. We nevertheless interpret Eq. (8) as descriptive rather than causal; low pooled correlation bounds collinearity, not confounding.

Against IQFormer-inspired, the fitted overlap coefficient is 14.38 percentage points per unit overlap and the SIR coefficient is -1.036 percentage points per dB. Magnitude skill is scored against an explicitly defined trivial baseline: the constant predictor outputs the mean gain over the 32 fitting cells (-8.27 percentage points for IQFormer-inspired, -4.63 percentage points for MCLDNN) and is evaluated unchanged on the 80 held-regime cells, with $R^2_{\text{skill}} = 1 - \text{MSE}_{\text{envelope}}/\text{MSE}_{\text{const}}$. On that basis the envelope attains RMSE 6.23 percentage points against 8.37 percentage points for the constant predictor, for $R^2_{\text{skill}} = 0.45$ (Fig. 3); the MCLDNN envelope gives 8.10 percentage points against 9.63 percentage points ($R^2_{\text{skill}} = 0.29$). Adding the cell-level mean SNR as a third covariate lowers the held-regime RMSE only modestly (to 6.15 and 8.05 percentage points, $R^2_{\text{skill}} = 0.46$ and 0.30), so the two-variable model remains primary. A covariate ladder on the same cells (full table in the Supporting Material) isolates each predictor: a SIR-only envelope attains held RMSE 6.84 percentage points against IQFormer-inspired (8.71 percentage points against MCLDNN) and an overlap-only envelope 7.46 percentage points (8.68 percentage points), versus 6.23 percentage points (8.10 percentage points) for the two-variable model. SIR is the stronger single covariate against IQFormer-inspired, whereas SIR and overlap have comparable single-covariate skill against MCLDNN. Adding overlap to SIR reduces held-regime RMSE by 0.61 percentage points against both references; a block-paired bootstrap over the held cells gives positive paired squared-error reductions with $P(\Delta\text{MSE} > 0) = 0.94$ and 0.85, but both 95% paired intervals include zero (Supporting Material), so we treat overlap as a consistent exploratory refinement beyond SIR rather than a separately resolved predictor. As a model-selection rule, the envelope mainly localizes the small region where the compact model becomes competitive rather than improving the average held-domain policy: the overlap–SIR sign rule selects A5 in 5 of the 80 held cells; among the 3 hindsight-optimal cells it identifies all three (recall $3/3$) with two extra selections (precision $3/5$), at 0.36 percentage points of oracle regret. Relative to always using IQFormer-inspired, this trades approximately 0.22 percentage points of realized macro-F1 for a 5.5% reduction in expected MACs. The 80 held cells span SIR from -10 to $+10$ dB; the -15-dB stratum occurs only in the hard-interference fitting split, so the held-regime transport metrics do not independently validate the break-even thresholds reported at -15 dB, although three held cells at SIR ≥ -10 dB still favor A5 in hindsight. A pooled oracle constant equal to the held-domain mean attains 6.04 and 6.81 percentage points, lower than the corresponding envelope RMSEs of 6.23 and 8.10 percentage points(marginally for IQFormer-inspired and more noticeably for MCLDNN). Because this oracle uses the held-domain mean in hindsight, the pooled gap does not indicate missing within-regime structure. Regime-wise results are heterogeneous: the envelope substantially outperforms the corresponding regime-specific constant on the unseen-speed and held-channel shifts, but not on the unseen-jammer and combined-OOD shifts. The envelope therefore provides regime-dependent transport and within-regime structure rather than uniform held-domain superiority. Sign agreement is 0.975 (0.950 for MCLDNN); with 0.963 (0.900) of held-regime gains negative, it is a secondary diagnostic only.

Solving Eq. (8) for zero gain gives break-even overlaps of 0.11 (vs. IQFormer-inspired) and 0.06 (vs. MCLDNN) at SIR $= -15$ dB, rising to 0.47 and 0.29 at SIR $= -10$ dB. The

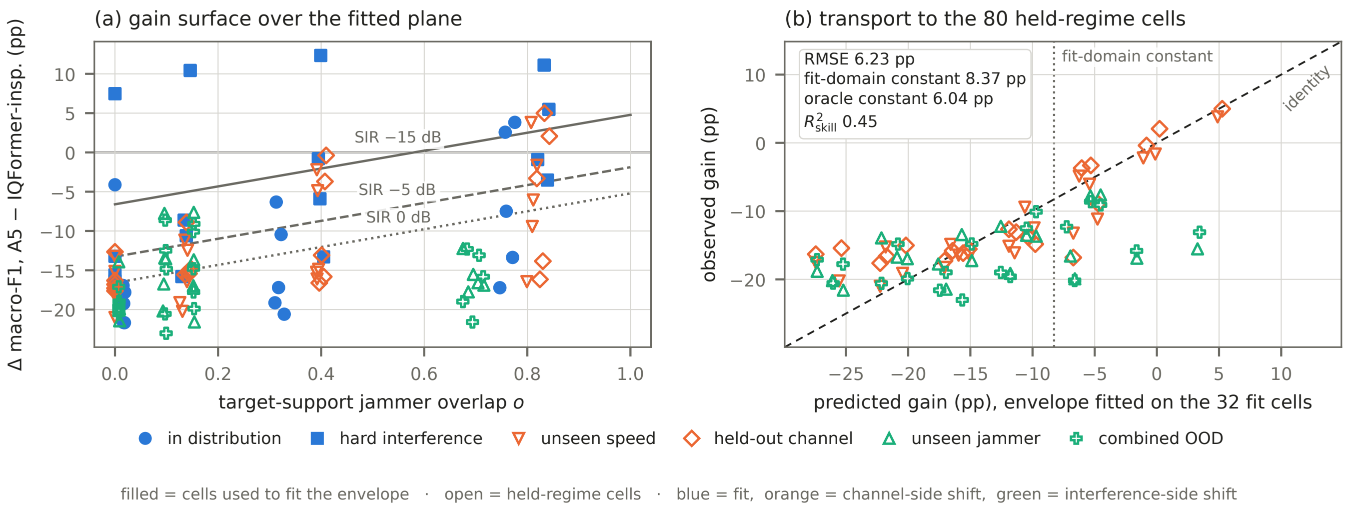

Fig. 3. Exploratory overlap–SIR operating envelope. The abscissa of panel (a) is the target-support jammer overlap of Eq. (9), not an occupied-bandwidth fraction. Filled blue markers are the cells that fit Eq. (8); open markers are held-regime cells, coloured by which side of the campaign is held out, orange for the channel-side splits (unseen speed, held-out channel) and green for the interference-side splits (unseen jammer, combined OOD). Panel (b) compares predicted and observed held-regime gains. RMSE is 6.23 percentage points versus 8.37 percentage points for a constant predictor estimated from the fitting cells ($R^2_{\mathrm{skill}} = 0.45$); an oracle constant equal to the held-regime mean would attain 6.04 percentage points.

TABLE III

HELD-REGIME ENVELOPE SKILL BY REGIME. GAINS AND RMSE IN PERCENTAGE POINTS; THE CONSTANT BASELINE IS THE MEAN GAIN OVER THE 32 FITTING CELLS. ONLY FINITE-SIR INTERFERED HELD CELLS ENTER THIS TABLE. OBSERVED GAIN IS THE UNWEIGHTED MEAN OVER THE 20 ENVELOPE CELLS OF EACH REGIME ON THE TEN-SEED AGGREGATE; IT IS NOT THE SAME STATISTIC AS THE SPLIT-LEVEL MACRO-F1 REPORTED ON THE MATCHED FIVE-SEED CAPACITY SUBSET.

Reference	Held regime	Cells	Obs. gain	Env. RMSE	Const. RMSE	R^2_{skill}
IQFormer-insp.	unseen speed	20	-12.4	3.94	7.87	0.75
IQFormer-insp.	held channel	20	-11.3	4.86	7.59	0.59
IQFormer-insp.	combined OOD	20	-16.5	7.50	9.23	0.34
IQFormer-insp.	unseen jammer	20	-16.1	7.74	8.67	0.20
IQFormer-insp.	pooled	80	-14.1	6.23	8.37	0.45
MCLDNN	unseen speed	20	-8.9	4.66	8.88	0.72
MCLDNN	held channel	20	-8.0	5.41	9.06	0.64
MCLDNN	combined OOD	20	-14.6	10.23	10.28	0.01
MCLDNN	unseen jammer	20	-14.4	10.34	10.24	-0.02
MCLDNN	pooled	80	-11.5	8.10	9.63	0.29

required overlap rises as interference becomes less severe, and these values are simulator-domain reference thresholds conditional on an overlap estimate, not deployment-ready control thresholds.

The preregistered mechanism expectation was non-increasing gain with overlap; the sealed directional test rejected that sign. Exploratory quintiles instead show monotonically increasing A5–CSSL gain, from a negative low-overlap region to a strongly positive high-overlap region; we return to this reversal in Section VI-C. The positive overlap direction is not specific to this adaptation reference: the fitted overlap coefficients against IQFormer-inspired and MCLDNN are also positive, at $+14.38$ and $+19.16$ percentage points per unit overlap, respectively.

D. Complexity and Complementarity

A5 remains useful outside its winning region because its errors differ from those of the I/Q models: late probability fusion improves over the stronger constituent by 4.71 percentage points with IQFormer-inspired and 5.70 percentage points with MCLDNN. The corresponding same-model seed-ensemble controls improve by only 1.49 and 1.69 percentage points, so the cross-model fusion gains exceed the ensemble controls by 3.22 and 4.01 percentage points. This is supporting evidence of complementary decisions, not marginal superiority.

Table I gives the corresponding parameter and MAC budgets; compactness is stated as a parameter and MACs property rather than an end-to-end inference-cost claim.

E. Clean-Boundary Diagnosis

The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 5, where a star denotes a class whose jammer-free condition was not seen during training. Among the 36 uncovered constellation-order class–model cells (nine compact-spectral models and three I/Q references, each over the three constellation-order classes QPSK, 16QAM, and 64QAM), transfer failure occurs in $27/27$ compact-spectral cells (100%) versus $1/9$ I/Q-reference cells (11%). The clean-retention gate did not pass. Figure 5 localizes the failure rather than hiding it in an aggregate.

We then followed a prospective decision chain. Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation; probe performance is the unweighted mean

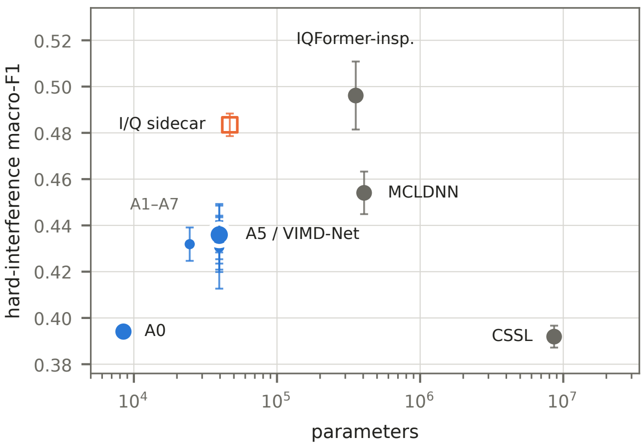

Fig. 4. Hard-interference macro-F1 versus parameter count on the matched five-seed subset; bars are ± 1 standard deviation over those seeds. Filled circles are sealed models (blue: the A0–A7 spectral family, of which only A0 and A5 are labelled; gray: the I/Q comparators); the open orange square is the prospective Tier-2 received-I/Q sidecar, which is exploratory and outside the sealed family. The Pareto reading is descriptive and regime-specific.

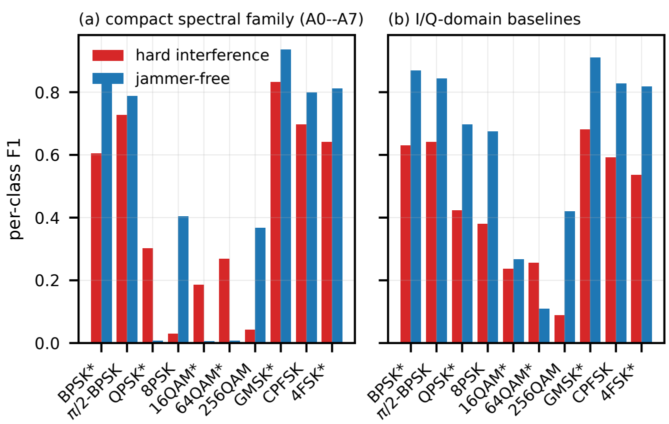

Fig. 5. Clean condition-transfer taxonomy for the compact spectral family and I/Q-domain baselines. A star marks a modulation class whose jammer-free condition was not observed during training; the four covered classes are unstarred. The plot localizes the clean boundary rather than aggregating it.

F1 over QPSK, 16QAM, and 64QAM (chance ≈ 0.1 on ten classes). The preregistered decision rule declares a representation *information-present, head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise, requiring *all* design–seed cells to exceed the threshold. For A5, five of the six design–seed cells (three algorithm seeds $\times$ the reconstructed jammer-free counterfactual and clean cross-validation probes) lie below the 0.25 threshold; the sole cell above it is borderline at 0.256, so we record the representation-limited side. The decision is unchanged at a threshold of 0.30 but flips at 0.20. The strong I/Q baselines exceed the threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, saturating near the modulated route; this arm is a negative control, not standalone evidence that a functional mask fallback was learned.

F. Prospective I/Q Repair and Capacity Controls

The next intervention adds a received-I/Q sidecar while retaining the spectral routes and the same source mixture. It increases the *total* model size from 39500 to 46794 parameters (+18%) and the MAC count to 43.1 million (about +3%). It raises clean macro-F1 by 11.77 percentage points ($[11.06, 12.49]$) and hard macro-F1 by 4.37 percentage points ($[3.44, 5.31]$), i.e. clean macro-F1 rises from 0.5005 to 0.6182 and hard-interference macro-F1 from 0.4406 to 0.4843. Its overall hard score is statistically near IQFormer-inspired ($\Delta = -0.29$ percentage points, $[-1.95, 1.46]$), while at severe SIR (-15 dB) it remains ahead of IQFormer-inspired by 12.09 percentage points ($[8.32, 15.88]$). The sidecar was designed prospectively after the clean-boundary diagnosis; it is interpreted as an intervention on the hypothesized information bottleneck, not as retroactive evidence for the original A5 architecture and not as a new multimodal fusion architecture. This substantially narrows the clean-condition failure while preserving the structured-interference advantage.

The sidecar is a two-channel depthwise-separable 1-D I/Q branch (2$\rightarrow$12$\rightarrow$12 depthwise$\rightarrow$24$\rightarrow$24 depthwise channels) with group normalization and SiLU activations, whose per-window temporal mean/std/max pooling is projected to a 24-dimensional embedding and fused residually with the spectral embedding (layer-level detail in the Supporting Material). The fused embedding drives a 10-class linear head whose logits are added residually to the spectral classifier logits under the shared training objective; the side head has no separate auxiliary supervision. It consumes only the received mixture and requires no component bookkeeping at inference.

Widening the spectral model without changing its representation domain separates capacity from representation. On the matched five-seed subset, hard-interference macro-F1 rises monotonically across the ladder, from 0.436 at the sealed S width to 0.455 at M (+1.94 percentage points) and 0.468 at L (+3.18 percentage points over S). The widest tier nonetheless stays below IQFormer-inspired on the marginal hard split (0.468 versus 0.496; -2.84 percentage points, $[-4.30, -1.31]$), while at SIR $= -15$ dB it retains the compact-model advantage (+14.07 percentage points, $[9.55, 18.51]$) and on clean retention it is still 18.1 percentage points behind that comparator. Width therefore raises the average and leaves the condition-dependent boundary in place; it does not substitute for the representation change made by the sidecar.

VI. Discussion

A. What the Operating Envelope Establishes

The evidence supports a practical model-selection statement within the simulator domain: the compact spectral configuration becomes relatively most competitive when interference is severe and jammer energy strongly overlaps the target-dominant time–frequency support, without confirming any specific routing or allocation mechanism (no sealed contrast isolates the teacher or route component; Section V-A). A larger I/Q model is preferable in weak, clean, or low-overlap conditions unless the compact system includes an I/Q side channel. The weak overlap–SIR correlation in the fitted cells bounds collinearity but does not establish causal independence.

B. What It Does Not Establish

Eq. (8) is empirical and descriptive, not a causal structural equation. The target-support overlap is simulator-derived because it requires separately tracked target and jammer components, and it is not directly observable at an uncooperative receiver; the break-even overlaps are therefore simulator-domain reference values conditional on an overlap estimate. The severe SIR corner is an in-sample stratification, and the held-out evaluation covers regimes within the same frozen simulator campaign, not real-world out-of-distribution conditions. A future prospective campaign should randomize overlap and SIR more independently and test the envelope against a receiver-observable overlap proxy; over-the-air or public-recording validation [35] is a separate external-validity step, since public benchmarks do not expose the tracked target and jammer components the overlap covariate requires.

C. Meaning of the Failed Directional Hypothesis

The preregistered directional mechanism hypothesis was not supported: the compact model's relative gain does not decrease with increasing overlap. The surviving result is therefore empirical rather than mechanistic: target-support jammer overlap organizes the observed rank exchange, but its direction does not confirm the originally hypothesized masking mechanism.

D. Representation Repair and Practical Model Selection

The sequence of interventions matters: the frozen probe narrows the cause, the coverage arm rejects a cheap explanation, the I/Q sidecar recovers a substantial part of the predicted missing-information deficit, and the capacity ladder separates width from representation. The Tier-2 evidence is more consistent with a representation-information gap than with insufficient clean exposure alone; the sidecar is a prospective intervention on that hypothesis, not the new main classifier and not a multimodal-fusion novelty.

E. Vehicular Relevance of the Operating Boundary

At $f_c = 5.9$ GHz and the held-speed conditions (Section III-A), the envelope can be read as a receiver-side model-selection reference for compute-constrained ITS/V2X receivers: high overlap marks structured interference concentrated on the target-dominant support, where the compact front end is relatively most competitive. At the 180 and 250 km/h held speeds, the nominal coherence time is approximately 0.43 and 0.31 ms, only about 42% and 30% of the 1024-μs decision window, making the held-speed regime a substantially more intra-window nonstationary condition than the lower-speed training cases. The practical reading is conditional on an observable overlap estimator (Section VI-B); the boundary is a simulation-domain design rule, not a deployment-ready controller.

VII. Limitations and Reproducibility

The evidence has the following explicit limits. First, all results are from source-disjoint TR 38.901 TDL-profile simulations, not a complete vehicular geometry or a MAC-/network-layer V2X evaluation. Second, the teacher uses simulation-component bookkeeping and cannot supervise unreferenced recordings. Third, the target-support overlap covariate is simulator-derived because it requires separately tracked target and jammer components, and is not directly available at a receiver. Fourth, the original confirmatory family gate did not pass: only the complete A5–A0 contrast is a positive sealed result, and the clean-retention gate likewise did not pass. Fifth, the severe-interference, operating-envelope, fusion, condition-transfer, probe, sidecar, and capacity analyses are exploratory or post-hoc, with post-hoc status stated where applicable. Sixth, the public-benchmark check validates comparator implementation only; external validation of the envelope is not yet established, and both I/Q comparators are unified-retraining references rather than exact recipe reproductions. Seventh, all models share one 30-epoch budget with a common early-stopping rule, and the budget binds asymmetrically: IQFormer-inspired and CSSL run to the limit in all ten seeds, so comparisons against them retain a possible budget-sufficiency caveat, while MCLDNN early-stops in every seed well inside the budget (median best epoch 13, range 11–16), making an immediately binding epoch cap an unlikely explanation for the severe-region A5–MCLDNN advantage (9.74 percentage points). A5 itself also runs to the budget in all ten seeds, so the concern is not one-sided. Eighth, the strongest compact-model advantage occurs at SIR $= -15$ dB, a stratum absent from the 80 held-regime cells; in a leave-one-SIR-stratum-out refit the severe rank reversal is preserved against MCLDNN but not against IQFormer-inspired (Supporting Material), so the corresponding break-even overlaps are conditional extrapolations of the fitted envelope rather than independently held-regime-validated thresholds.

Reproducibility is enforced at the artifact boundary. Each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256, and every figure is rendered from a CSV file with its own provenance manifest; the resolved macros, figure-source CSVs, per-seed aggregates, run configurations, and the full provenance manifest are released as a reproduction package upon acceptance. The frozen run stores cache identity, configurations, checkpoints, source-aligned predictions, paired statistics, and execution records, preventing a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. Conclusion

Compact spectral and high-capacity I/Q representations exchange rank across the structured-interference plane. A low-dimensional overlap–SIR envelope transports the gain level across regime cells excluded from fitting with measurable skill relative to a fit-domain constant baseline, while a post-hoc severe-interference analysis identifies a reproducible compact-model advantage in one corner of the plane. The original sealed family resolves a positive whole-configuration A5–A0 contrast but not the isolated teacher or route-count effects. A prospective probe–coverage–sidecar chain further indicates that the clean-condition failure is repairable by restoring a small amount of received-I/Q information, while spectral capacity alone does not remove the boundary. The result is therefore a simulation-domain operating envelope and representation-repair study, not a claim of uniform superiority or a deployment-ready selector.

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